

Pre-class assignment

- 1) In Bayes' Theorem, what is the utility of including the prior and evidence probabilities?
(In other words, why is it useful to include them?)
- 2) What is the difference between the frequentist and Bayesian approaches toward statistical inference, and how does it manifest in the way these approaches are used?

DID THE SUN JUST EXPLODE?
(IT'S NIGHT, SO WE'RE NOT SURE.)

THIS NEUTRINO DETECTOR MEASURES
WHETHER THE SUN HAS GONE NOVA.

THEN, IT ROLLS TWO DICE. IF THEY
BOTH COME UP SIX, IT LIES TO US.
OTHERWISE, IT TELLS THE TRUTH.

LET'S TRY.
DETECTOR! HAS THE
SUN GONE NOVA?



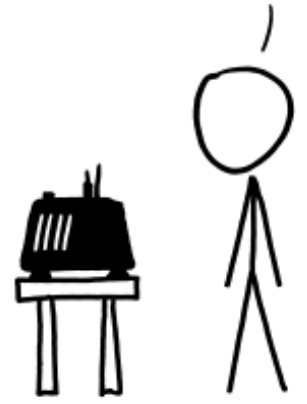
FREQUENTIST STATISTICIAN:

THE PROBABILITY OF THIS RESULT
HAPPENING BY CHANCE IS $\frac{1}{36} = 0.027$.
SINCE $p < 0.05$, I CONCLUDE
THAT THE SUN HAS EXPLODED.



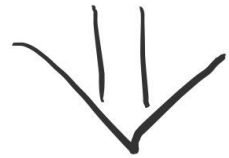
BAYESIAN STATISTICIAN:

BET YOU \$50
IT HASN'T.



Bayes' theorem:

$$P(x|y) = \frac{P(y|x) \times P(x)}{P(y)}$$



$$\left[\begin{array}{c} \text{probability of } x \\ \text{given } y \end{array} \right] = \frac{\left[\text{prob. of } y \text{ given } x \right] \times \left[\begin{array}{c} \text{prob. of} \\ x \end{array} \right]}{\left[\text{prob. of } y \right]}$$

or

$$\left[\text{posterior} \right] = \frac{\left[\text{Likelihood} \right] \times \left[\text{prior} \right]}{\left[\text{evidence} \right]}$$

In terms of a model (w/ params Θ) and data (d):

$$p(\Theta|d) = \frac{p(d|\Theta) p(\Theta)}{p(d)}$$

or

Diagrammatic representation of Bayes' theorem:

Left side (Posterior): $\left[\begin{array}{l} \text{probability that model} \\ \text{parameters } \Theta \text{ describe} \\ \text{this data } d \end{array} \right]$ (labeled posterior)

Right side (Likelihood and Prior): $\left[\begin{array}{l} \text{probability of} \\ \text{data given model} \\ \text{parameters} \end{array} \right] \times \left[\begin{array}{l} \text{prob of} \\ \text{model} \\ \text{parameters} \end{array} \right]$ (labeled Likelihood and prior)

Denominator (Evidence): $\left[\begin{array}{l} \text{probability of data} \end{array} \right]$ (labeled Evidence)

Equation: $\left[\begin{array}{l} \text{probability that model} \\ \text{parameters } \Theta \text{ describe} \\ \text{this data } d \end{array} \right] = \frac{\left[\begin{array}{l} \text{probability of} \\ \text{data given model} \\ \text{parameters} \end{array} \right] \times \left[\begin{array}{l} \text{prob of} \\ \text{model} \\ \text{parameters} \end{array} \right]}{\left[\begin{array}{l} \text{probability of data} \end{array} \right]}$

Question You receive a test for a certain disease. 2% of people in your demographic have the disease, and the test is 90% reliable. If the test is positive, how likely are you to actually have the disease?

Bayesian approach:

$$P(\text{disease given positive test}) = \frac{P(T|D) \times P(D)}{P(T)}$$

Diagram illustrating the Bayesian approach formula with red arrows indicating the components:

- $P(D|T)$ (red) points to $P(\text{disease given positive test})$
- $P(T|D)$ (red) points to $P(\text{test given data})$
- $P(D)$ (red) points to $P(\text{data})$
- $P(T)$ (red) points to $P(\text{test being positive})$

$$P(D|T) = \frac{P(T|D) P(D)}{P(T)}$$

$$P(T|D) = 0.9 = \text{accuracy of test}$$

$$P(D) = 0.02 = \text{chance of a person in your demo having this disease}$$

$$P(T) = (\text{prob. of true positive}) + (\text{prob. of false positive})$$

$$= P(T|D) P(D) + P(T|\text{not } D) P(\text{not } D)$$

$$= 0.9 \times 0.02 + (1 - 0.9) \times (1 - 0.02)$$

$$P(D|T) \approx 0.1552 \sim 15.5\%$$

Follow-up: how does the probability of you actually having the disease change as a function of test reliability? (Try reliability values of 0.8, 0.95, 0.99, 0.999)

reliability	P(DIT)
0.75	0.0577
0.8	0.0754
0.9	0.1552
0.95	0.2794
0.99	0.6689
0.999	0.9532
0.9999	0.9951
0.99999	0.999

Context: medical tests are essentially never 100% reliable, and some are more reliable than others. Example, flu antigen tests (rapid tests) are only moderately sensitive (50-70% sensitivity, but only 1-5% false positives). Flu molecular tests are much more sensitive (>90%, with <5% false positives).

Question: If I have photometric observations of a galaxy that might be a high-redshift galaxy but could be other things (e.g., a closer, red galaxy), how do I assess the Bayesian probability of the high- z hypothesis?

Some hints:

- 1) what do you know about galaxy evolution over time?
- 2) how do you include both true positives and false positives?

probability
of high- z hypothesis

Likelihood that
high- z galaxy model
matches data

prior based on
galaxy distribution as
 $f(\text{redshift})$

$$P(\Theta|d) = \frac{P(d|\Theta) P(\Theta)}{P(d)}$$

= $P(\text{prob. of data
matches to model})$

false positives

$$P(d) = P(\text{data fit to
high- z model}) P(\text{high } z
galaxies) + P(\text{data fit to
low- z model}) \times$$

true positives

$P(\text{low- z
galaxies})$

plugging in some (hand-heavy and unreliable) numbers:

$$p(d | \text{high } z) = 0.7 \quad [\text{high-redshift model matched to data}]$$

$$p(d | \text{low } z) = 0.4 \quad [\text{low-redshift model matched to data}]$$

$$p(\text{high } z \text{ gal}) = 0.1$$

$$p(\text{low } z \text{ gal}) = 0.9$$

} probability representing underlying galaxy distribution (more galaxies at late time!)

} note: there do not need to add up to 1. Why?

$$P(\text{high } z \text{ galaxy} | \text{data}) = \frac{0.7 \times 0.1}{0.7 \times 0.1 + 0.4 \times 0.9}$$

$$= 0.16279 \rightarrow \approx 16.3\% \text{ galaxy is at high redshift!}$$

notes:

- underlying sample matters: more galaxies at low redshift are observable! (so even a low prob. of error for a given galaxy impacts the outcome significantly)
- model uniqueness/data fit matters a lot: this is why few-band photo- z is awful, many-band helps w/ excluding false positives

photo- z
PDF for a $z \sim 6$ blue galaxy w/ 5 HST bands might look a lot like this:

